

B. The analytic identity

Given a candidate growth regime H_θ and the baseline transition kernel $F_0(v|u)$ from the fitted copula, the **analytic identity**

$$F_\theta(v) = \mathbb{E}_U[H_\theta(F_0(v|U))] = \frac{1}{n} \sum_{i=1}^n H_\theta(F_0(v|u_i))$$

predicts the current-grade CDF *exactly*—no Monte Carlo simulation is needed. The uniform regime $H(p) = p$ (grey dashed) systematically misses the observed CDF, while the inferred $H_{\hat{\theta}}$ (teal dashed) closes the gap.